

Modbus slave Map

Configuration name:	IMOX.uwp
Project name:	
Current date:	7/5/2026 2:44:32 PM
IDE version:	9.1.10.3.260220.082452
Modbus TCP/IP port:	502
Modbus RTU settings:	Disable

Name			
Address			
Zip code		City	
Country		Phone	
E-mail			
Web page			
Contact person			
Order information			

Modbus slave Map

[F1] Temperature (-> L1 Default L2 ACC_ROOM)

Name	Function	Address (Hex/Dec)	Format	Label	Multiplier	Min	Max	Invalid Value	Note Reference
Status									
Status	0x04 IR (Read)	0x0000 / 00000	Uint16		1				116
Flags	0x04 IR (Read)	0x0001 / 00001	Uint16		1				117
Total value	0x04 IR (Read)	0x0002 / 00002	Uint64		1	0	9999999999 9999		
Adjustable value	0x04 IR (Read)	0x0006 / 00006	Uint16		1	0	9999999999 9999		
Input value	0x04 IR (Read)	0x0007 / 00007	Int64		1	- 10000000000 00000	9999999999 9999		
Counter rollover value	0x04 IR (Read)	0x000B / 00011	Int16		1				
Command									
Action	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x000C / 00012	Uint16		1				118
Argument	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x000D / 00013	Uint64		1	0	9999999999 9999		
Enable rollover counter	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x0011 / 00017	Uint16		1				
Local calendar	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x0012 / 00018	Uint16		1				

[F2] Alarm (-> L1 Default L2 ACC_ROOM)

Modbus slave Map

Name	Function	Address (Hex/Dec)	Format	Label	Multiplier	Min	Max	Invalid Value	Note Reference
Status									
Status	0x04 IR (Read)	0x0028 / 00040	Uint16		1				100
Flags	0x04 IR (Read)	0x0029 / 00041	Uint32		1				101
Reset timer	0x04 IR (Read)	0x002B / 00043	Uint32	s	1	0	86399		
Siren timer	0x04 IR (Read)	0x002D / 00045	Uint32	s	1	0	86399		
Lamp status	0x04 IR (Read)	0x002F / 00047	Int16		1				
Alarm ON event	0x04 IR (Read)	0x0030 / 00048	Uint32		1	0	2147483647		
Alarm OFF event	0x04 IR (Read)	0x0032 / 00050	Uint32		1	0	2147483647		
Alarm Ack event	0x04 IR (Read)	0x0034 / 00052	Uint32		1	0	2147483647		
Action	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x0036 / 00054	Uint16		1				102
Reset timer	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x0037 / 00055	Uint32	s	1	0	86399		
Siren timer	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x0039 / 00057	Uint32	s	1	0	86399		
Acknowledgment with auto reset	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x003B / 00059	Uint16		1				
Siren ON when return to normal state	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x003C / 00060	Uint16		1				
Local calendar	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x003D / 00061	Uint16		1				

Modbus slave Map

[F3] Smoke Status (-> L1 Default L2 ACC_ROOM)

Name	Function	Address (Hex/Dec)	Format	Label	Multiplier	Min	Max	Invalid Value	Note Reference
Status									
Status	0x04 IR (Read)	0x0060 / 00096	Uint16		1				116
Flags	0x04 IR (Read)	0x0061 / 00097	Uint16		1				117
Total value	0x04 IR (Read)	0x0062 / 00098	Uint64		1	0	9999999999 9999		
Adjustable value	0x04 IR (Read)	0x0066 / 00102	Uint64		1	0	9999999999 9999		
Input value	0x04 IR (Read)	0x006A / 00106	Int64		1	- 10000000000 00000	9999999999 9999		
Counter rollover value	0x04 IR (Read)	0x006E / 00110	Int16		1				
Command									
Action	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x006F / 00111	Uint16		1				118
Argument	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x0070 / 00112	Uint64		1	0	9999999999 9999		
Enable rollover counter	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x0074 / 00116	Uint16		1				
Local calendar	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x0075 / 00117	Uint16		1				

[F5] Zone temperature (-> L1 Default L2 ACC_ROOM)

Modbus slave Map

Name	Function	Address (Hex/Dec)	Format	Label	Multiplier	Min	Max	Invalid Value	Note Reference
Status									
Status	0x04 IR (Read)	0x0088 / 00136	Uint16		1				153
Flags	0x04 IR (Read)	0x0089 / 00137	Uint16		1				154
Ambient temperature	0x04 IR (Read)	0x008A / 00138	Int16	°C	10	-500	1000		
Disable timer	0x04 IR (Read)	0x008E / 00142	Uint32	s	1	0	86399		
Command									
Action	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x0090 / 00144	Uint32		1				155
Disable	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x0092 / 00146	Uint16		1				156
Heating set point selection	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x0093 / 00147	Int16		1				157
Set Heating SP1	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x0094 / 00148	Int16	°C	10				
Set Heating SP2	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x0095 / 00149	Int16	°C	10				
Set Heating SP3	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x0096 / 00150	Int16	°C	10				
Add offset to Heating set points	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x0097 / 00151	Int16	°C	10	-500	1000		

Modbus slave Map

Heating fan speed mode	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x0098 / 00152	Int16		1				158
Cooling set point selection	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x0099 / 00153	Int16		1				157
Set Cooling SP1	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x009A / 00154	Int16	°C	10				
Set Cooling SP2	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x009B / 00155	Int16	°C	10				
Set Cooling SP3	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x009C / 00156	Int16	°C	10				
Add offset to Cooling set points	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x009D / 00157	Int16	°C	10	-500	1000		
Cooling fan speed mode	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x009E / 00158	Int16		1				158
Heating function									
Status	0x04 IR (Read)	0x009F / 00159	UInt16		1				159
Flags	0x04 IR (Read)	0x00A0 / 00160	UInt16		1				160
Set point selected	0x04 IR (Read)	0x00A1 / 00161	Int16		1				161
Temperature	0x04 IR (Read)	0x00A2 / 00162	Int16	°C	10	-500	1000		
Set point	0x04 IR (Read)	0x00A3 / 00163	Int16	°C	10	-200	2500		
PID value	0x04 IR (Read)	0x00A4 / 00164	UInt16	%	1	0	100		
Safe ON timer	0x04 IR (Read)	0x00A5 / 00165	UInt32	s	1	0	86399		
Safe OFF timer	0x04 IR (Read)	0x00A7 / 00167	UInt32	s	1	0	86399		
Heating SP1									

Modbus slave Map

Heating SP1	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x00B2 / 00178	Int16	°C	10	-200	2500		
Minimum value	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x00B3 / 00179	Int16	°C	10	-200	2500		
Maximum value	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x00B4 / 00180	Int16	°C	10	-200	2500		
Dead band	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x00B5 / 00181	Int16	°C	10	-100	100		
Heating SP2									
Heating SP2	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x00B6 / 00182	Int16	°C	10	-200	2500		
Minimum value	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x00B7 / 00183	Int16	°C	10	-200	2500		
Maximum value	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x00B8 / 00184	Int16	°C	10	-200	2500		
Dead band	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x00B9 / 00185	Int16	°C	10	-100	100		
Heating control mode									

Modbus slave Map

Control mode	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x00BE / 00190	Uint16		1	1	2		164
Cycle time	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x00BF / 00191	Uint16	s	1	1	3600		
Hysteresis	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x00C0 / 00192	Uint16	°C	10	0	100		
Proportional band	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x00C1 / 00193	Uint16	°C	10	10	100		
Proportional coefficient	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x00C2 / 00194	Uint16	%	1	0	100		
Integrative coefficient	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x00C3 / 00195	Uint16	%	1	0	100		
Maximum integrative error	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x00C4 / 00196	Uint16	°C	10	10	100		
Derivative coefficient	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x00C5 / 00197	Uint16	%	1	0	100		
Minimum output value	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x00C6 / 00198	Uint16	%	1	0	100		

Modbus slave Map

Maximum output value	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x00C7 / 00199	Uint16	%	1	0	100		
Default output value	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x00C8 / 00200	Uint16	%	1	0	100		
Safe mode	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x00C9 / 00201	Uint16		1	1	3		167
ON time	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x00CA / 00202	Uint16	min	1	1	60		
OFF time	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x00CB / 00203	Uint16	min	1	1	60		
Antifreeze set point	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x00CC / 00204	Uint16	°C	10	10	600		
Disable automation timer	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x0113 / 00275	Uint32	s	1	0	86399		
Local calendar	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x0115 / 00277	Uint16		1				

Modbus slave Map

[F29] PLC - SIEMENS (-> L1 Default)

Name	Function	Address (Hex/Dec)	Format	Label	Multiplier	Min	Max	Invalid Value	Note Reference
Status									
Status	0x04 IR (Read)	0x0294 / 00660	Uint16		1				116
Flags	0x04 IR (Read)	0x0295 / 00661	Uint16		1				117
Total value	0x04 IR (Read)	0x0296 / 00662	Uint16		1	0	9999999999 9999		
Adjustable value	0x04 IR (Read)	0x0297 / 00663	Uint16		1	0	9999999999 9999		
Input value	0x04 IR (Read)	0x0298 / 00664	Uint16		1	- 10000000000 00000	9999999999 9999		
Counter rollover value	0x04 IR (Read)	0x0299 / 00665	Int16		1				
Command									
Action	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x029A / 00666	Uint16		1				118
Argument	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x029B / 00667	Uint16		1	0	9999999999 9999		
Enable rollover counter	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x029C / 00668	Uint16		1				
Local calendar	0x03 HR (Read) 0x06 HR (Write) 0x10 HR (Write)	0x029D / 00669	Uint16		1				

Note reference legend

Note 100: Alarm - State values

State=01: Alarm OFF
State=02: Alarm ON
State=03: Alarm was ON
State=04: Acknowledged, alarm ON
State=05: Acknowledged, alarm was ON
State=06: Reset alarm

Note 101: Alarm - Flag bit status

Bit=00: Internal use (Alarm is OFF)
Bit=01: Alarm is ON
Bit=02: Alarm is ON, input was ON
Bit=03: Alarm is acknowledged
Bit=04: Alarm is acknowledged, input was ON
Bit=08: Input is ON
Bit=09: Internal use (Acknowledged)
Bit=10: Internal use (Siren ON)
Bit=11: Reset is active
Bit=12: Test input: force ON
Bit=13: Test input: force OFF
Bit=14: Reset is active, reset timer is running
Bit=15: Siren is active, siren timer is running
Bit=16: Internal use (Ack request)
Bit=17: Internal use (Auto reset)
Bit=18: Internal use (Siren latched)
Bit=19: Internal use (Siren timeout)
Bit=20: Internal use (Pulse reset)

Note 102: Alarm - Action values

Action=00:
Action=01: Alarm acknowledgement
Action=02: Reset (ignore timer)
Action=03: Reset ON
Action=04: Reset ON with timer
Action=05: Reset OFF
Action=06: Reset ON/OFF toggle
Action=07: Reset ON with timer/OFF toggle
Action=08: Test alarm signal ON
Action=09: Remove test alarm signal ON
Action=10: Test alarm signal ON (activate/deactivate)
Action=11: Test alarm signal OFF
Action=12: Remove test alarm signal OFF
Action=13: Test alarm signal OFF (activate/deactivate)

Note 116: Counter - State values

State=01: Enabled (adjustable value unfrozen)
State=02: Disabled (adjustable value frozen)

Modbus slave Map

Note 117: Counter - Flag values

Bit=00: Freeze

Note 118: Counter - Action values

Action=00:

Action=01: Increase adjustable value

Action=02: Decrease adjustable value

Action=03: Set adjustable value

Action=04: Reset adjustable value (set default value)

Action=05: Freeze adjustable value

Action=06: Unfreeze adjustable value

Action=07: Freeze/Unfreeze adjustable value toggle

Action=08: Reset rollover counter value

Command actions 01, 02, 03 should be combined with the register "Command argument".

Example: increase by two the total value of the counter function

Set Modbus register "Command argument" to 0x02

Set Modbus register "Command Action" to 0x01

Note 153: Zone temperature - State values

State=01: Not controlling

State=02: Controlling heating

State=03: Controlling cooling

State=04: Controlling

State=05: Not controlling (Disable activated)

State=06: Controlling heating (Disable activated)

State=07: Controlling cooling (Disable activated)

State=08: Controlling (Disable activated)

Note 154: Zone temperature - Flag bit status

Bit=00: Heating control is running

Bit=01: Cooling control is running

Bit=02: Automation disable is ON

Bit=03: No localization available for: Function.TemperatureZone.Livelcon.TimerDisable

Note 155: Zone temperature - Action bit commands

Bit=00: Heating activation

Bit=01: Heating deactivation

Bit=02: Heating toggle activation / deactivation

Bit=03: Heating set point selection

Bit=04: Set Heating SP1

Bit=05: Set Heating SP2

Bit=06: Set Heating SP3

Bit=07: Add offset to Heating set points

Bit=08: Heating fan speed mode

Bit=09: Activate Heating Force ON

Bit=10: Deactivate Heating Force ON

Modbus slave Map

Bit=11: Heating force ON toggle activation / deactivation
Bit=12: Activate Heating Force OFF
Bit=13: Deactivate Heating Force OFF
Bit=14: Heating force OFF toggle activation / deactivation
Bit=15: Cooling activation
Bit=16: Cooling deactivation
Bit=17: Cooling toggle activation / deactivation
Bit=18: Cooling set point selection
Bit=19: Set Cooling SP1
Bit=20: Set Cooling SP2
Bit=21: Set Cooling SP3
Bit=22: Add offset to Cooling set points
Bit=23: Cooling fan speed mode
Bit=24: Activate Cooling Force ON
Bit=25: Deactivate Cooling Force ON
Bit=26: Cooling force ON toggle activation / deactivation
Bit=27: Activate Cooling Force OFF
Bit=28: Deactivate Cooling Force OFF
Bit=29: Cooling force OFF toggle activation / deactivation

It's possible to execute multiple action commands at the same time by writing consecutive HR registers (function code 16); if the registers to be changed are not consecutive, it's necessary to send two commands as shown in the examples below.

Example 1: select heating set point OFF

Write 1 in the Modbus register "command heating setpoint selection" (Function code 6)
Write 8 (bit 3) in the Modbus register "command action" (Function code 16)

Example 2: select heating set point 2 (HSP2)

Write 3 in the Modbus register "command heating setpoint selection" (Function code 6)
Write 8 (bit 3) in the Modbus register "command action" (Function code 16)

Example 3: set value to heating set point 1 (set 21,5 °C)

Write 215 in the Modbus register "command set heating setpoint 1" (Function code 6)
Write 16 (bit 4) in the Modbus register "command action" (Function code 16)

Example 4: set value to heating set point 2 (set 19 °C)

Write 190 in the Modbus register "command set heating setpoint 2" (Function code 6)
Write 32 (bit 5) in the Modbus register "command action" (Function code 16)

Example 5: add 2°C offset to all heating setpoints

Write 20 in the Modbus register "command add offset to heating setpoints" (Function code 6)
Write 128 (bit 7) in the Modbus register "command action" (Function code 16)

Example 5: Select speed AUTO for fan coil (heating)

Write 1 in the Modbus register "command heating fan speed mode" (Function code 6)
Write 256 (bit 8) in the Modbus register "command action" (Function code 16)

Example 6: Activate force ON (heating)

Write 512 (bit 9) in the Modbus register "command action" (Function code 16)

Example 7: Deactivate force ON (heating)

Write 1024 (bit 10) in the Modbus register "command action" (Function code 16)

Modbus slave Map

Note 156: Zone temperature - Disable automation commands

Disable=01: Disable ON

Disable=02: Disable ON + Timeout

Disable=03: Disable OFF

Disable=04: Disable ON/OFF Toggle

Disable=05: Disable ON/OFF Toggle + Timeout

Example 1: disable the automations for the temperature zone function

Write 2 in Modbus register "Command Disable" (Function code 6)

Example 2: toggle disable/enable automations for the temperature zone function

Write 4 in Modbus register "Command Disable" (Function code 6)

Note 157: Zone temperature - Activate Set point commands

Set=01: OFF

Set=02: S1

Set=03: S2

Set=04: S3

Note 158: Zone temperature - Fan coil commands

Mode=01: Auto

Mode=02: Speed 1

Mode=03: Speed 2

Mode=04: Speed 3

Mode=05: OFF

Note 159: Zone temperature - Control state values

State=01: Control is OFF

State=02: Controlling Set Point 1 - Status is OFF

State=03: Controlling Set Point 1 - Status is ON

State=04: Controlling Set Point 2 - Status is OFF

State=05: Controlling Set Point 2 - Status is ON

State=06: Controlling Set Point 3 - Status is OFF

State=07: Controlling Set Point 3 - Status is ON

State=08: Controlling manual set point - Status is OFF

State=09: Controlling manual set point - Status is ON

State=10: Safe mode is active forcing OFF status

State=11: Safe mode is active forcing ON status

State=12: Antifreeze Control is forcing ON status

State=13: Auxiliary Control is forcing ON status

State=14: ON status is forced

State=15: Antifreeze Control is forcing ON status

State=16: OFF status is forced

State=17: System Function Control is forcing OFF status

Note 160: Zone temperature - Control flag values

Bit=00: Set Point OFF

Bit=01: Set Point S1

Modbus slave Map

Bit=02: Set Point S2
Bit=03: Set Point S3
Bit=04: Set Point manual
Bit=05: Safe mode is active - OFF state
Bit=06: Safe mode is active - ON state
Bit=07: Control Status in ON
Bit=08: Antifreeze Control is forcing ON state
Bit=09: Auxiliary Control is forcing ON
Bit=10: Forced ON
Bit=11: Auxiliary Control is forcing OFF
Bit=12: Forced OFF
Bit=13: System Function Control is forcing OFF

Note 161: Zone temperature - Set Point Selected

State=01: OFF
State=02: SP1
State=03: SP2
State=04: SP3
State=05: Manual

Note 164: Zone temperature - Control Mode values

Mode=01: Comparator
Mode=02: PID

Note 167: Zone temperature - Safe Mode values

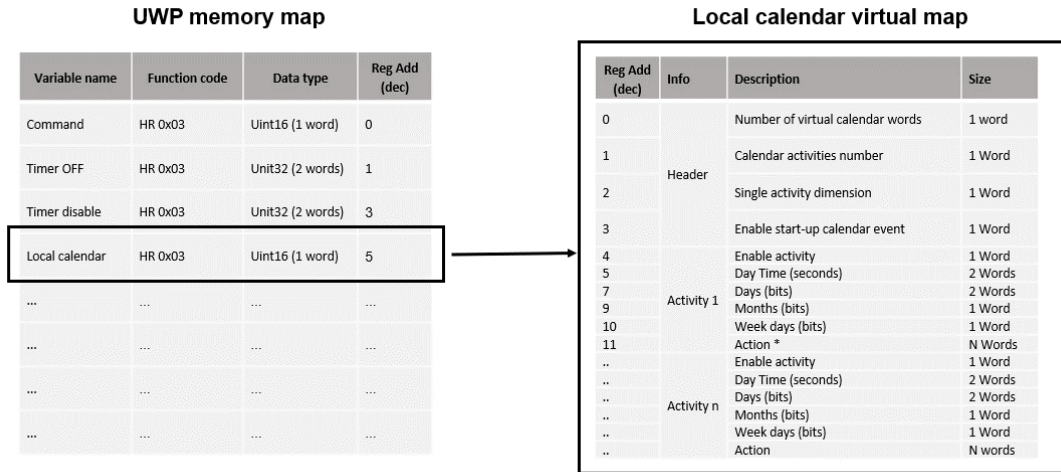
Value=01: None
Value=02: Turn output OFF
Value=03: Recycle output

Modbus slave Map

Local calendar settings

The mapping of the calendar activities of each function is carried out through a virtual memory, which is reserved to each function with a/including local calendar.

Each function has a dedicated local calendar register that points to a different virtual memory area from the other functions: the memory area size varies according to the number of activities defined in the local calendar and according to the type of function where the local calendar is used.



Reading commands notes:

1. The starting address it is the real Modbus map one (1 word)
2. By reading one single word from the starting address of the real Modbus map, you will get the total dimension of the virtual map (total number of words for all the local calendar activities)
3. It is not necessary to read all the activities at the same time (you can read N words from the starting address and you will have a partial overview of the virtual map)

Writing commands notes:

1. The starting address it is the real Modbus map one
2. The number of words to be written must be the total dimension of the virtual map (you have always to write the whole map)
3. By writing a single word starting from the Start address of the calendar of the real Modbus map, it is possible to set on/off the “Enable calendar events at start-up” bit of the first 16 activities of the calendar.

Words order for the virtual map:

Register address	Info	Description	Size
0	Header	Number of virtual calendar words	1 word
1		Number of activities	1 Word
2		Single activity dimension	1 Word
3		Enable start-up calendar event	1 Word
4	Activity 1	Enable activity	1 Word
5		Day Time (seconds)	2 Words
7		Days (bits)	2 Words
9		Months (bits)	1 Word
10		Week days (bits)	1 Word

Modbus slave Map

11		Action**	N Words
..			
..	Activity n	Enable activity	1 Word
..		Day Time (seconds)	2 Words
..		Days (bits)	2 Words
..		Months (bits)	1 Word
..		Week days (bits)	1 Word
..		Action**	N words

* *The action register indicates the action to be performed and depends on the function (for example, for a Counter function, the action register consists of 2 words; for a switch function instead, the Action register consist of 1 single word).

Reserved registers

The Modbus map provides a reserved register that contains information about the UWP system status.

Function code	Address (hex/dec)	Format	Value	UWP status description
0x03 HR 0x04 IR	0xFFFF / 65535		0x0101	- Ongoing configuration (busy) - System start-up (busy)
0x03 HR 0x04 IR	0xFFFF / 65535		0x0000	Runtime (ready)